A1.2 Data representation and computer logic

Subject: Computer science Unit: A1 Computer fundamentals Grade: 11DP SL

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 Data representation 60 minutes	A1.2.1–A1.2.2 Convert non-negative integers between binary, decimal and hexadecimal, including binary ↔ hexadecimal. Explain binary encoding and how integers, characters/strings, images, audio and video are stored and retrieved.	30 min: model place value and 4-bit hexadecimal groups; guided conversion practice. 20 min: match data types to encoding examples: character codes, pixels, sampled sound and video frames/audio. 10 min exit check: convert 26 to binary/hex and explain one encoding mechanism. Support: place-value grid and code table. Extend: explain how representation choices affect storage. Homework: six conversions and one encoding explanation.	IB guide p. 29, A1.2.1–A1.2.2; teacher conversion worksheet; character table; pixel grid; sampled-wave diagram. Parent support: ask students to explain one worked conversion.
Lesson 2 Gates and circuits 60 minutes	A1.2.3–A1.2.5 Describe the purpose of logic gates in binary computing; recognise standard AND, OR, NOT, NAND, NOR, XOR and XNOR symbols. Construct truth tables and connect circuit diagrams, Boolean expressions and problem descriptions.	20 min: retrieve conversions; model seven gates, their functions and one control-system application. 30 min: pairs draw a two-input circuit, label intermediate outputs and construct its full truth table; match it to an expression/scenario. 10 min exit check: distinguish OR/XOR and analyse a combined circuit. Support: gate reference sheet and partially completed table. Extend: construct a circuit from a problem description. Homework: two combined-circuit truth tables.	IB guide p. 29, A1.2.3–A1.2.5; standard gate-symbol sheet; truth-table worksheet; logic simulator such as CircuitVerse: https://circuitverse.org/

Lesson 3 Simplification and application 60 minutes	A1.2.4–A1.2.5 Use Boolean algebra rules and Karnaugh maps to simplify output expressions. Construct equivalent logic diagrams and verify that simplification preserves outputs.	15 min: Boolean identities and worked algebraic simplification. 20 min: guided Karnaugh map from a truth table; demonstrate adjacent grouping with a manageable example. 20 min: exam-style task linking circuit → truth table → expression → simplified circuit. 5 min exit check: justify equivalence and identify remaining difficulty. Support: Boolean-rule sheet and pre-labelled map. Extend: solve a three-input example independently. Use results to decide whether further guided practice is needed.	IB guide p. 29, A1.2.4–A1.2.5; Boolean identities sheet; Karnaugh-map templates; teacher exam-style questions and worked answers; circuit simulator.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	This is a compressed first pass. Review exit checks after each lesson. If combined circuits remain insecure, reteach them before simplification. Use short homework to consolidate; schedule additional practice if needed. A1.2 is common SL/HL content; no separate HL lesson is required here.		

Technology Integration دمج التكنولوجيا	Use a conversion demonstration, pixel grid and character-code examples. Students predict circuit outputs before checking them in a simulator.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: positional notation, powers of two, logical equivalence and algebra. Physics: digital signals and logic circuits. Media: digital image/audio representation.
External Assessment Integration دمج التقييمات الخارجية	Prepare for Paper 1 through describe/explain responses, conversions, truth tables and circuit construction. Use teacher-created guide-aligned exam-style questions; review reasoning and intermediate steps.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: trace and simplify. Communication: explain a conversion/circuit to a partner. Self-management: complete brief practice and reflect on exit-check errors. Normalise debugging and allow private thinking time.
Standards المعايير	IB Computer science guide, first assessment 2027, p. 29: A1.2.1–A1.2.5. All five objectives introduced across three sessions; mastery checked formatively.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	TOK: How does representing information in binary influence what is preserved or lost? Can two different circuit diagrams represent the same knowledge? Support answers with encoding examples or matching truth tables.
Success Criteria معايير النجاح (Measurable)	MUST: Convert an integer between binary/decimal/hex; explain one binary encoding; recognise all seven gates and complete a simple truth table. SHOULD: Explain encoding across text, images, audio and video; construct a combined circuit and its expression/table; simplify with a supplied rule sheet or map. COULD: Independently simplify a three-input example and verify equivalent outputs. These levels guide support, not exemptions from syllabus content.
National Identity Domain مجالات الربط بالهوية الوطنية	Context: use a UAE smart-building sensor scenario to express an alarm condition and construct its logic circuit.
	Values: discuss responsible handling of digitally encoded personal information and accurate technical communication.
	Application: students explain how logic-based controls can support efficient lighting and resource use in UAE buildings; use a hypothetical scenario rather than claiming specific national performance data.